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Structure of the respiratory system

The respiratory system is made up of air passages through the nose, a tube called the wind-pipe, or **trachea**, and the lungs. The lungs are situated in a space inside the chest called the **thoracic cavity**. The walls of this space are supported by the ribs and its floor is a sheet of muscle called the **diaphragm** (Fig. 1).

The inner surface of the thoracic cavity and the outer surface of the lungs are covered with a shiny slippery skin called the **pleural membrane** (Fig. 1). It produces **pleural fluid**. This fluid acts as a lubricant, reducing friction as the lungs rub against the ribs during breathing.

Air breathed in through the nose is warmed, made moist, and has dust and germs removed from it before it reaches the lungs. The air passages down to the lungs are lined with a carpet of microscopic hairs called **cilia** (Fig. 2C). Between the cilia are cells which produce sticky liquid called **mucus** in which germs and dust are trapped. Cilia wave back and forth like oars, moving the mucus with trapped germs and dust up to the throat, where it is swallowed.

The wind-pipe has walls supported by curved strips of cartilage (Fig. 1). These hold the wind-pipe open even when the neck is bent. The voice box, or **larynx**, is at the top of the trachea. Membranes called **vocal cords** are attached to the sides of the voice box. Normally these membranes are slack and air passes soundlessly between them. But when muscles pull them taut and air is forced between them, they vibrate and make sounds which form the voice.

At its base the wind-pipe divides into two tubes called **bronchi** (Figs. 1 and 2). One bronchus leads into each lung. Inside the lungs the bronchi divide like the branches of a tree into extremely narrow tubes called **bronchioles**. The smallest bronchioles have muscles in their walls which can make them wider or narrower, thereby altering the rate of air flow in and out of the lungs.

Bronchioles end in tiny bubble-like air sacs called **alveoli** (Fig. 2B). Each alveolus is about 0·2 mm in diameter and there are about 300 million in a set of human lungs. Alveoli give the lungs a spongy texture. A network of capillaries surrounds each alveolus (Fig. 2B). The blood in these capillaries absorbs oxygen from air breathed into the alveoli, and releases carbon dioxide into the air breathed out of the alveoli.

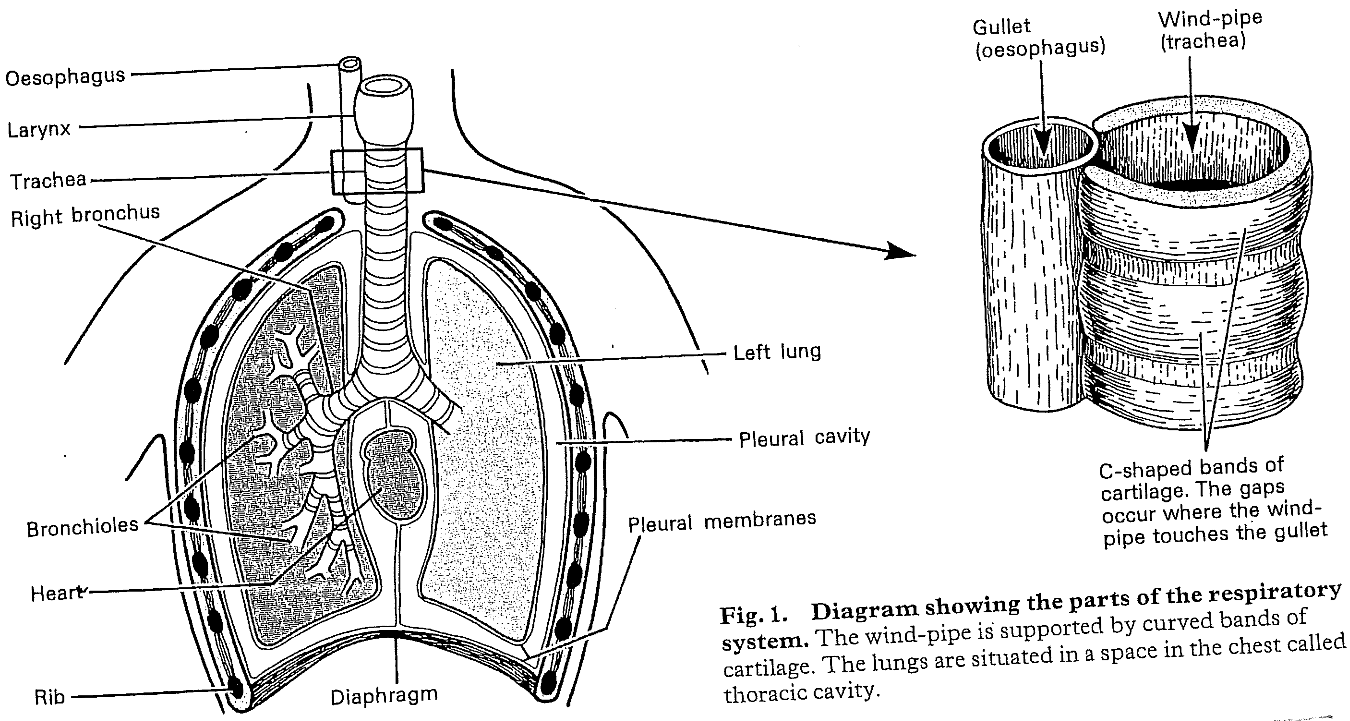
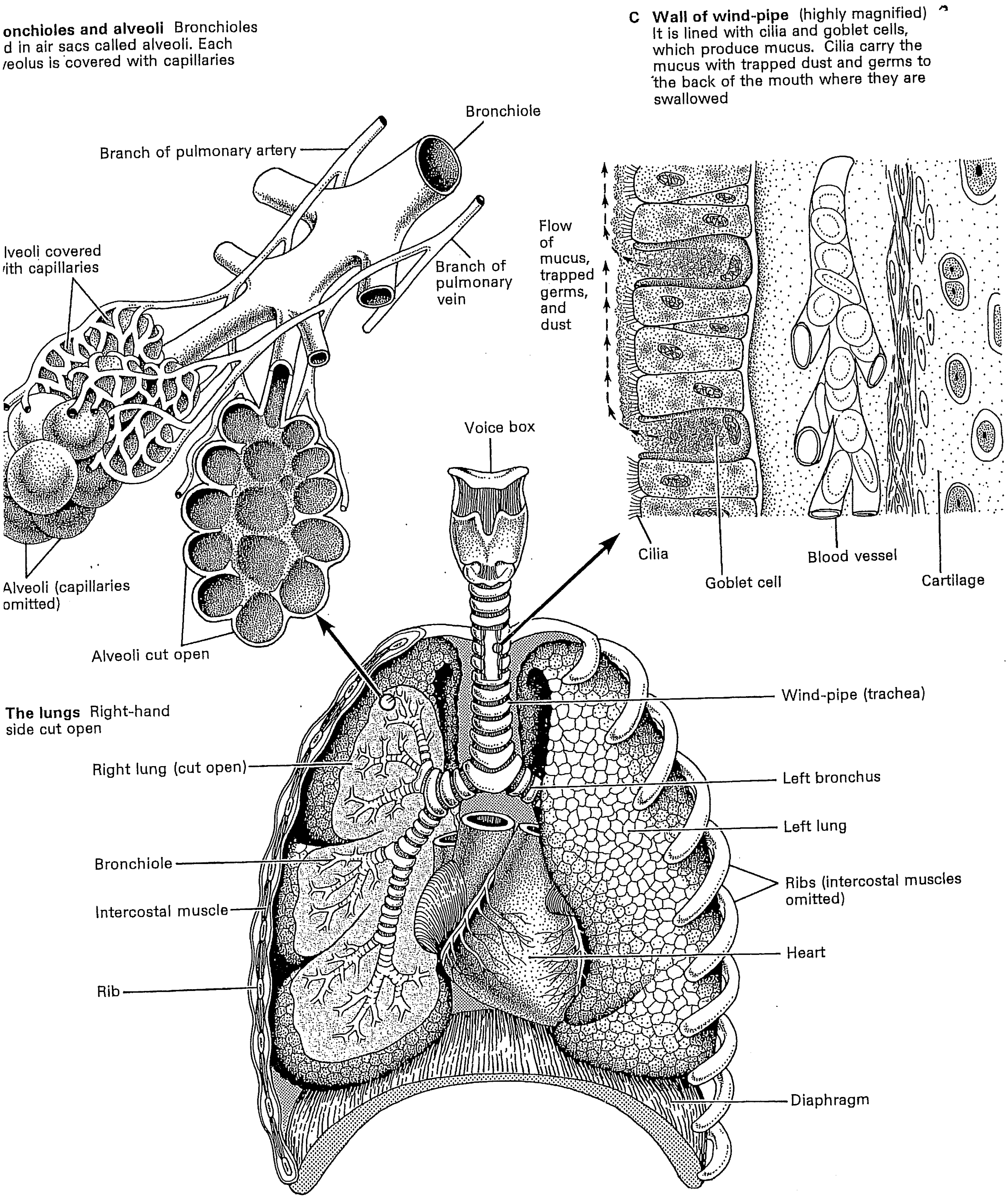


Fig. 1. Diagram showing the parts of the respiratory system. The wind-pipe is supported by curved bands of cartilage. The lungs are situated in a space in the chest called t thoracic cavity.



'ig. 2. **Structure of the respiratory system.**

2

Breathing

Air is moved in and out of the lungs by the action of the **diaphragm** and **intercostal muscles.**

Inspiration (breathing in)

The diaphragm is dome-shaped and relaxed before a breath is taken (Fig. 1B). During the first stage of breathing in, the diaphragm **contracts** and becomes flatter in shape (Fig. 1A). At the same time **external intercostal muscles** (Fig. 4) contract, pulling the ribs upwards so that they pivot where they join the backbone (Fig. 3). The flattening of the diaphragm and the lifting of the rib cage increase the volume of the thoracic cavity. This is automatically followed by an increase in lung volume. The increase in volume reduces air pressure inside the lungs, and air rushes into them from the atmosphere through the air passages.

Expiration (breathing out)

The diaphragm and external intercostal muscles relax. The rib cage drops and the diaphragm returns to its original dome shape. These movements reduce the volume of the thoracic cavity and the lungs return to their original volume. This squeezes air out of the lungs through the air passages. Air can be forced out of the lungs by contracting the **internal intercostal muscles.** This happens, for example, when playing a wind instrument.

At rest, an adult breathes about 16–18 times a minutes, using the diaphragm alone. During exercise this rate increases, and the intercostal muscles are used to increase the volume of each breath. This supplies the muscles with extra oxygen and removes carbon dioxide more quickly.

Figure 2 demonstrates how the intercostal muscles work. Fit an elastic band through holes A and B and watch how the length changes as you move the apparatus. The elastic band represents the external intercostal muscles (which raise the rib cage). Then fit the elastic band through holes C and D. It now represents the internal intercostal muscles (which lower the rib cage, forcing air from the lungs).

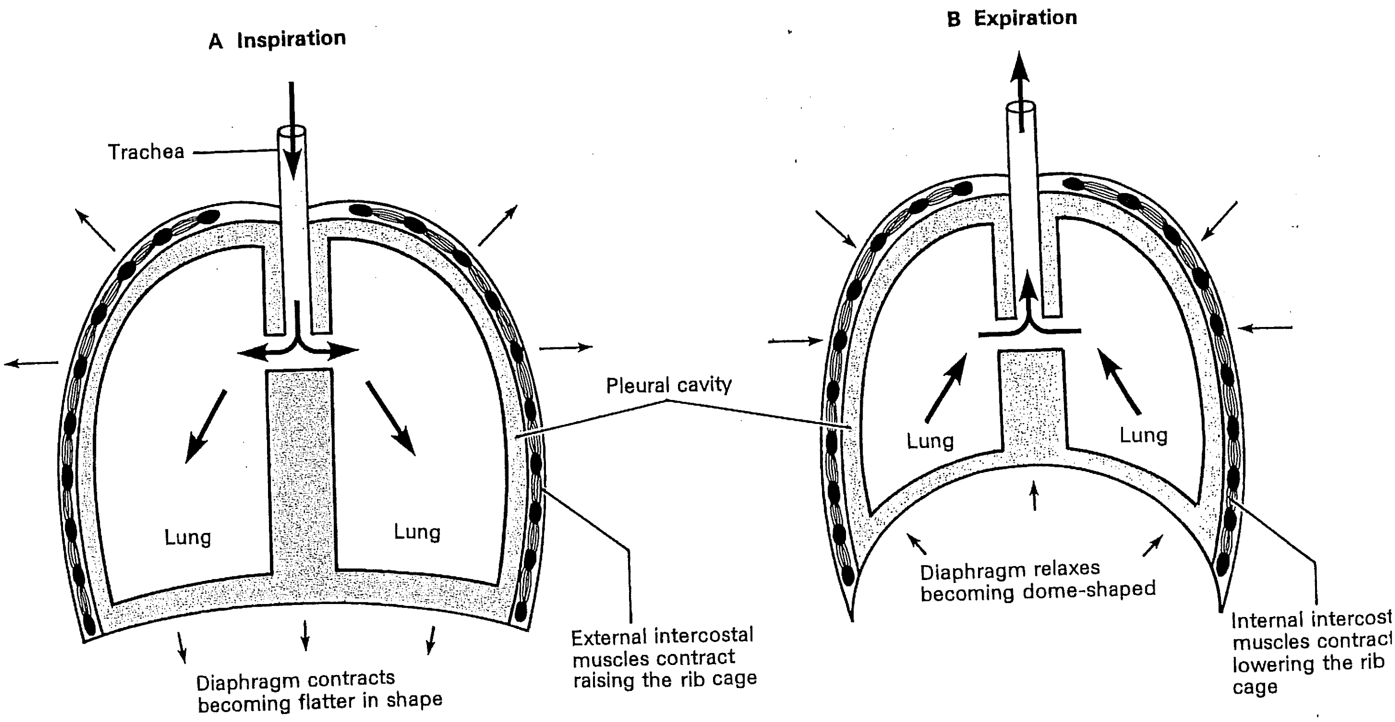


Fig. 1. Diagram of breathing movements. A Contraction of the intercostal muscles and diaphragm increases the volume of the rib cage inflating the lungs. **B** Relaxation of these muscles reduces the volume of the rib cage, squeezing air out of the lungs.

3

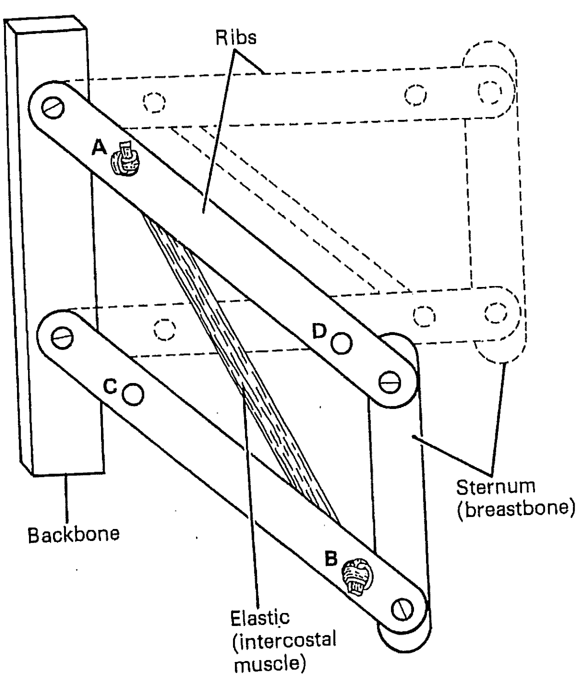


Fig. 2. Model showing how intercostal muscles move the rib cage.

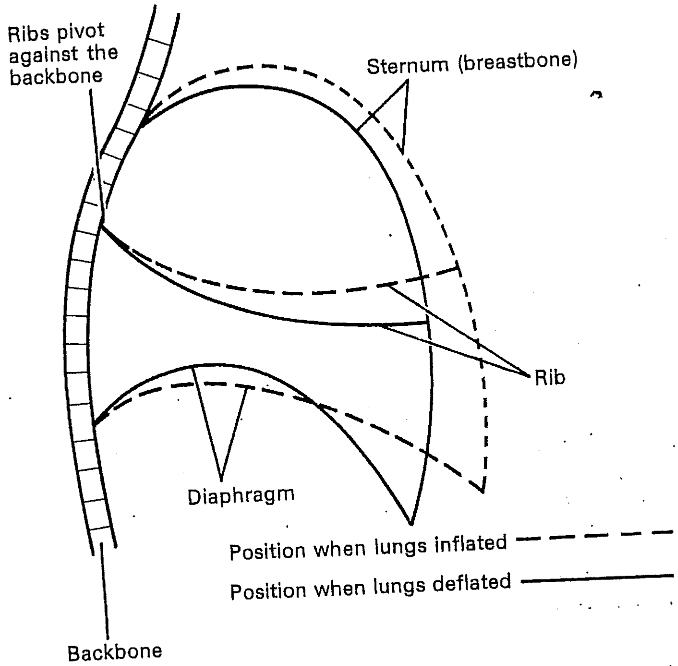


Fig. 3. Side view of the rib cage showing movements during breathing.

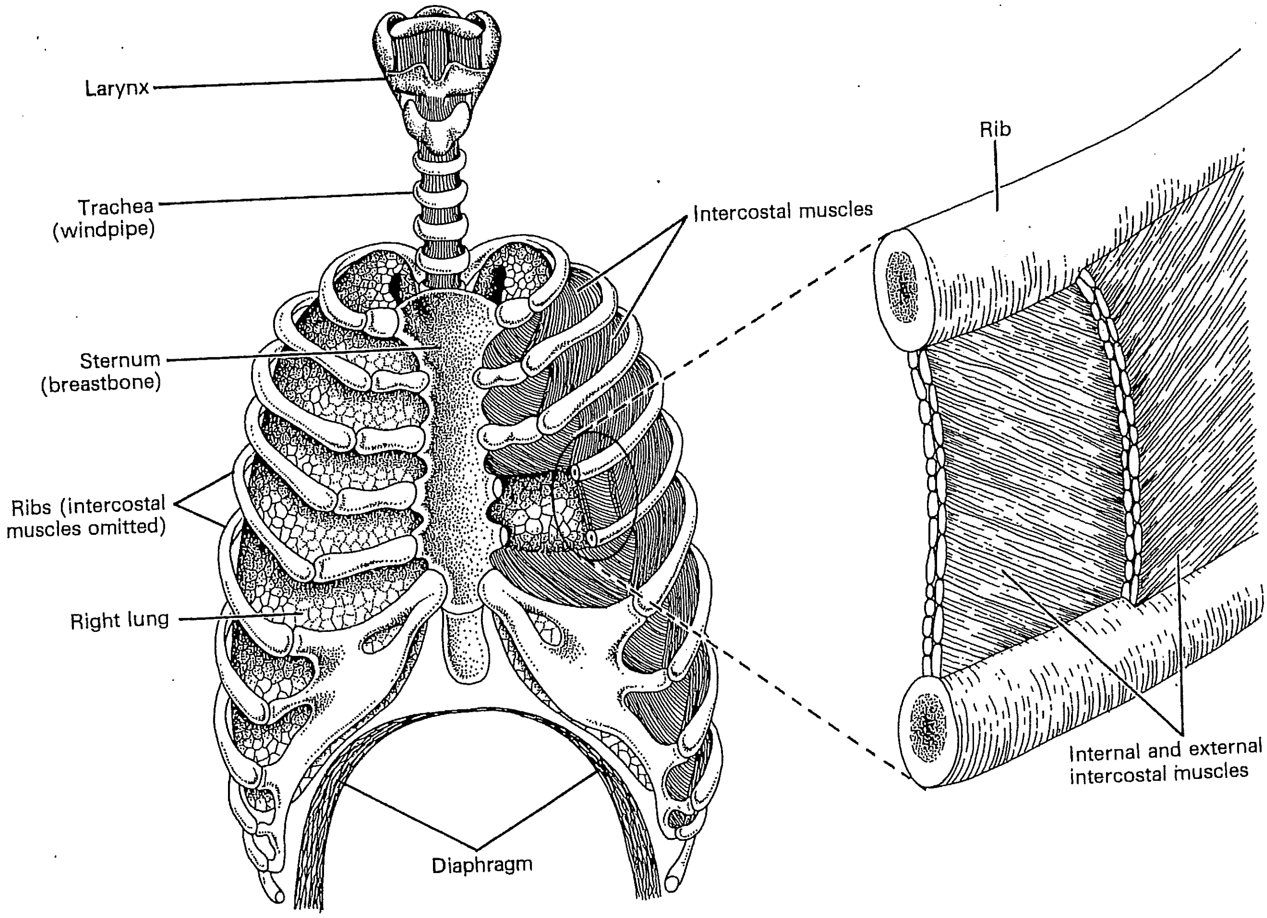


Fig. 4. Rib cage, showing the position of intercostal muscles.

4

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Gaseous exchange

The body absorbs oxygen from the air and at the same time it releases into the air carbon dioxide produced by respiration. In other words the body 'exchanges gases' with the atmosphere. The technical name for this process is **gaseous exchange,** and it takes place within the alveoli of the lungs.

Absorption of oxygen

Blood entering the capillaries of the lungs is **deoxygenated**; that is, it contains no oxygen. This is because haemoglobin in its red cells has released all its oxygen to the cells of the body. But air breathed from the atmosphere into the alveoli is rich in oxygen. This oxygen dissolves in a film of water lining the alveoli (Figs. 1 and 2B), then diffuses through the alveoli and capillary walls, a distance of only 0·005 mm, into the blood. Here, it combines with haemoglobin in the red cells making oxyhaemoglobin.

The capillaries that cover the alveoli walls are narrower in diameter than red blood cells. The red cells are squeezed out of shape as they are forced through the lung capillaries by blood pressure. This helps oxygen absorption in two ways. First, it causes a large area of each red cell to be pressed against capillary walls through which oxygen is diffusing. Second, red cells are slowed down as they squeeze through the capillaries, thus increasing the time available for oxygen absorption. By the time blood leaves the lung capillaries it is oxygenated.

Release of carbon dioxide

Blood entering the lung capillaries is full of carbon dioxide which it has collected from respiring cells on its journey around the body. Most of the carbon dioxide is dissolved in the plasma either in the form of **sodium bicarbonate** or **carbonic acid.** A little carbon dioxide is transported by red cells as a substance called **carbamino-haemoglobin.**

As blood enters the lungs an enzyme in the red cells causes chemical reactions which break down sodium bicarbonate and carbamino-haemoglobin, releasing carbon dioxide gas. This gas diffuses through the capillary and alveoli walls into the film of water and then into the alveoli. Finally, it is removed from the lungs during breathing out.

Differences between breathed and unbreathed air

	Unbreathed air	*Breathed air from a sleeping man*	*Breathed air from a running man*
Nitrogen	78%	78%	78%
Inert gases	1%	1%	1%
Oxygen	21%	17%	12%
Carbon dioxide	Trace	4%	9%
Water vapour	Variable	Saturated	Saturated

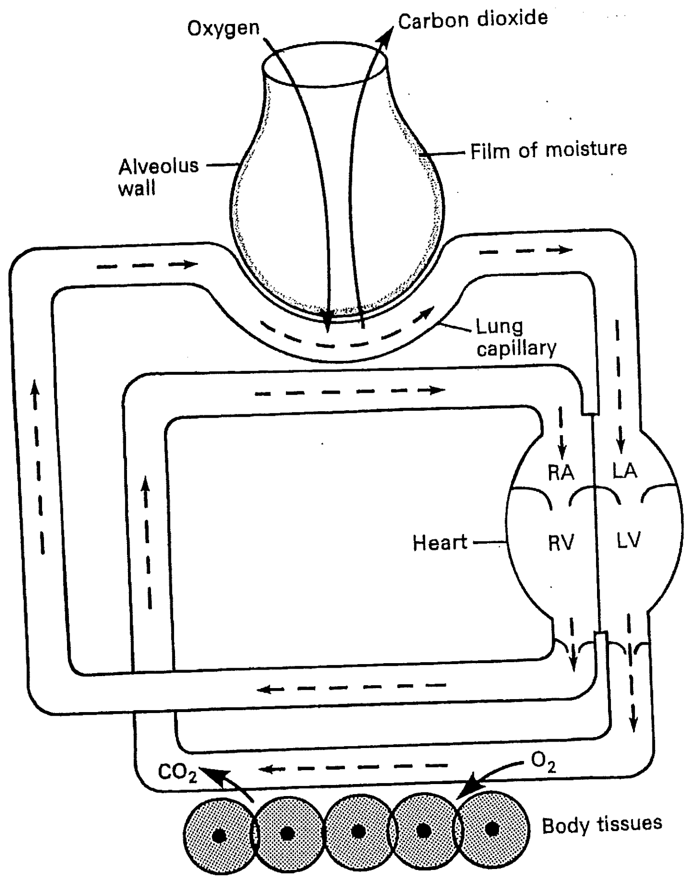
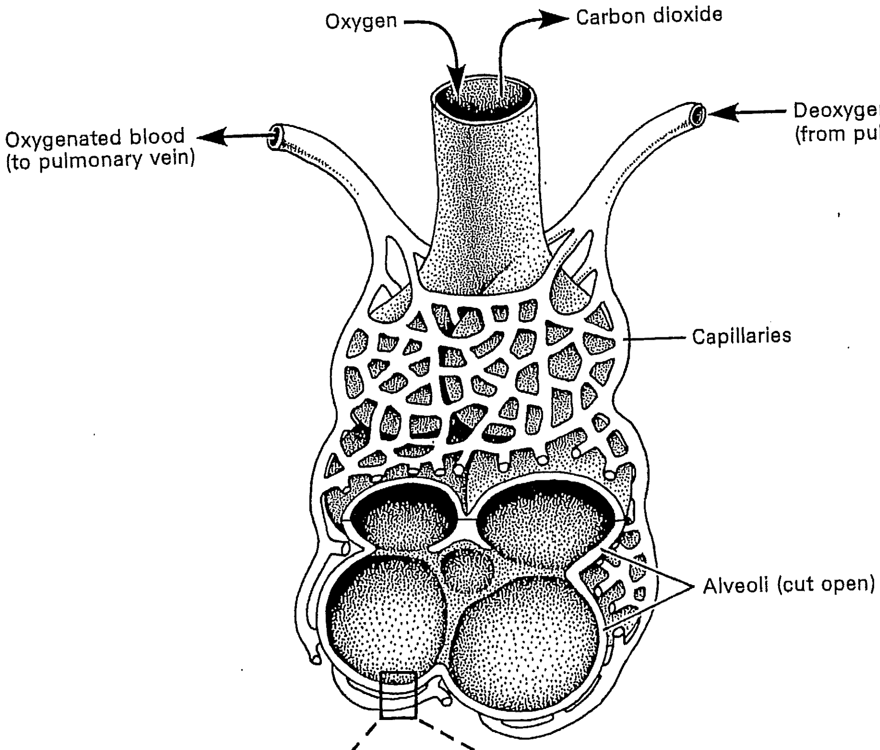


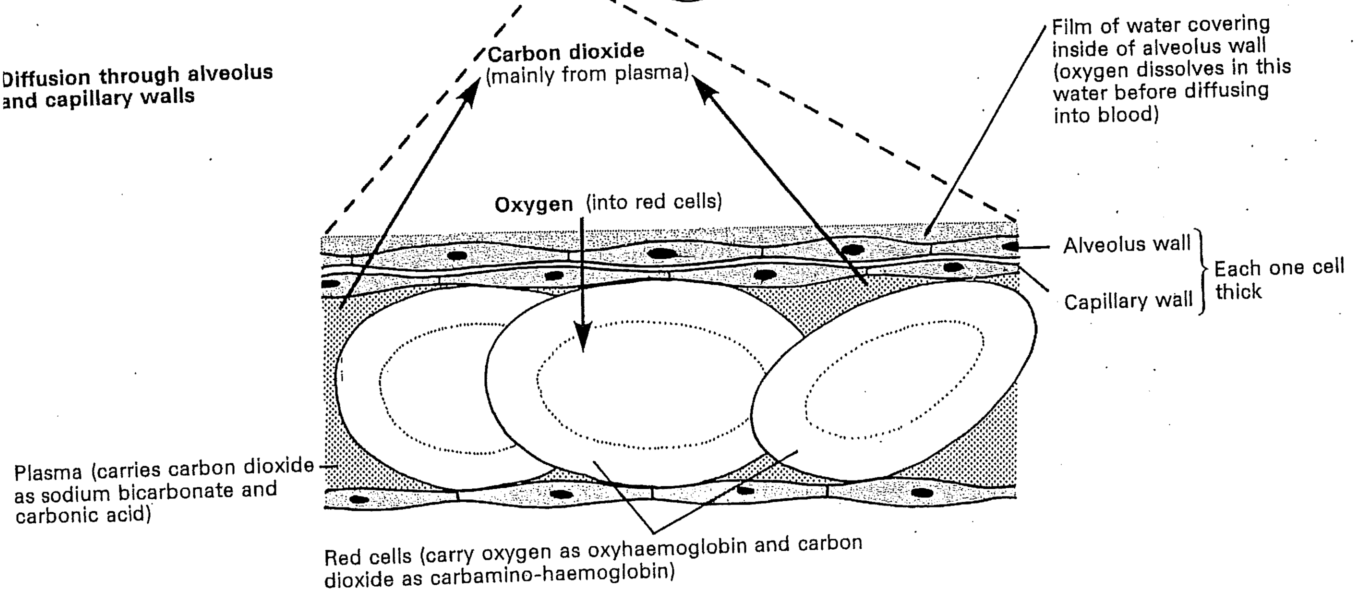
Fig. 1. Diagram of gaseous exchange (shown in more deta opposite).

5

group of alveoli cut open



Diffusion through alveolus and capillary walls



ig. 2. **Gaseous exchange** is the absorption of oxygen from ıe air 'in exchange' for carbon dioxide which is released from ıe body. This exchange takes place in the alveoli of the lungs. ›xygen breathed into the lungs diffuses through a film of ıoisture lining the alveoli, and then passes through the alveoli and capillary walls into the red cells. Here it forms oxyhaemoglobin. Carbon dioxide is carried by blood mainly as sodium bicarbonate. An enzyme in red cells breaks this down releasing carbon dioxide gas. This diffuses into the alveoli and is breathed out of the lungs.

6

Respiratory System Questions

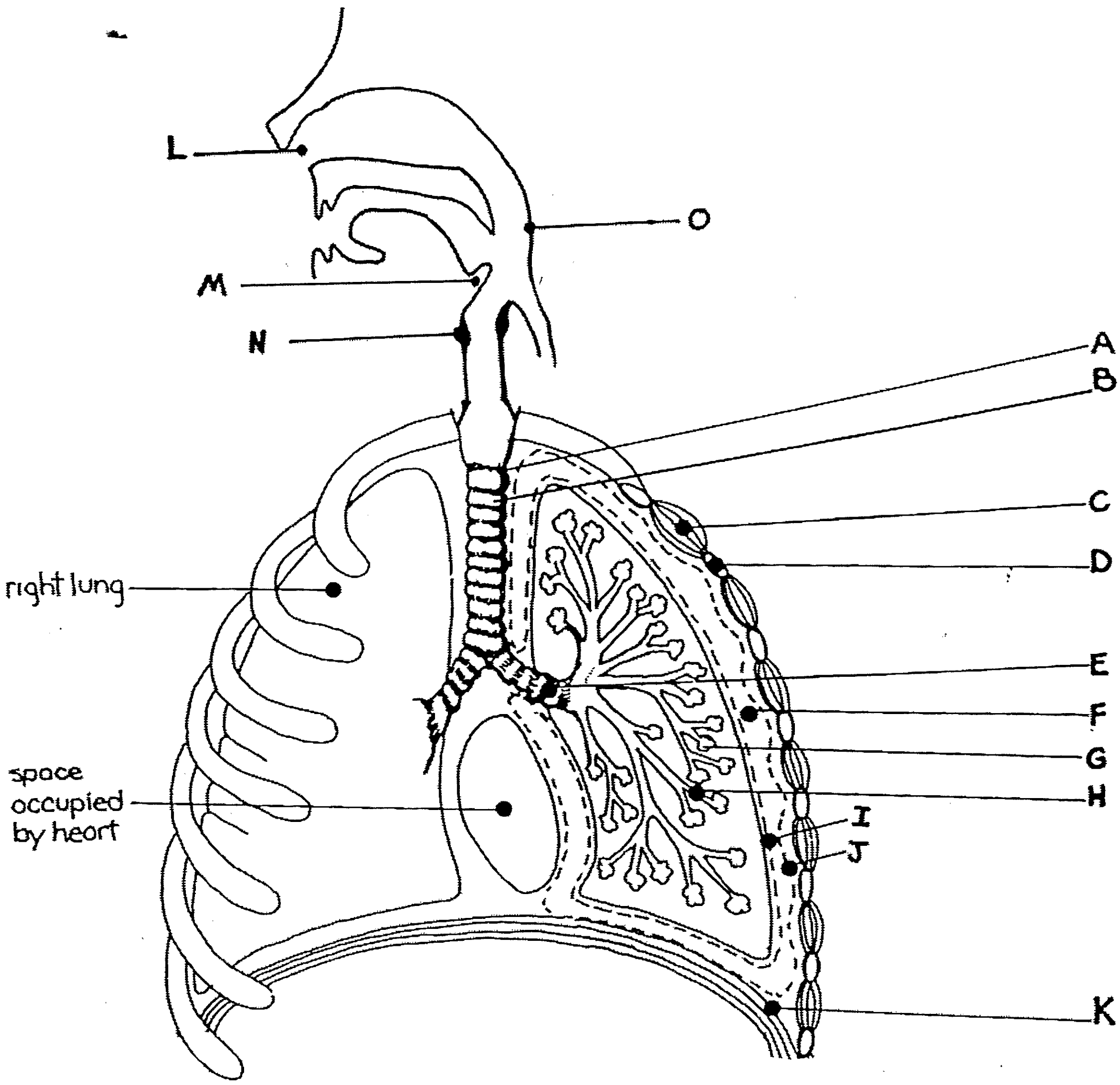
1. Why do multi-cellular organisms require a system of organs and tissues to enable the exchange of gases between the environment and cells?

____

____

____

2. Label the following diagram of the human respiratory system.



3. a) Use a pen/pencil to draw arrows representing the flow of air from the nose/mouth to an alveolus on question 2.

b) List the name of the structures that the air passes through to reach the alveoli.

____

____

____

____

7

4. Explain the structure and function of the following.

Structures	Structure/function
Tracheal Cartilage	
Trachea	
Intercostal muscle	
Rib	
Left bronchus	
Pleural Cavity	
Alveoli	
Bronchiole	
Pleural membranes	
Diaphragm	
Nasal cavity	
Epiglottis	
Pharynx	
Larynx	
Lung	

5. Define the following terms;

a) Breathing – ______________________________

____

b) Inspiration – ______________________________

____

c) Expiration – ______________________________

____

d) Cellular respiration – ______________________________

____

8

6. Fill in the missing words to complete the following explanation of the 'mechanics' of inspiration.
   Thoracic cavity increases by
   a) Intercostal muscles ______________________.
   b) Ribs ___________ and move _____________.
   c) Diagram _______________ and moves _______________.
   d) Sternum moves _____________.
   e) This leads to the volume of the thoracic cavity _______________ thus _______________ the air pressure in the lungs causing air to move ____________ the lungs.

7. Fill in the missing words to complete the following explanation of the 'mechanics' of inspiration.
   Thoracic cavity increases by
   a) Intercostal muscles ______________________.
   b) Ribs ___________ and move _____________.
   c) Diaphragm _______________ and moves _______________.
   d) Sternum moves _____________.
   e) This leads to the volume of the thoracic cavity _______________ thus _______________ the air pressure in the lungs causing air to move ____________ the lungs.

8. The concentration of which gas in the blood causes a change in the breathing rate?

____

9. Define the following terms;
   a) Diffusion – ____

   ____

   b) Diffusion gradient. – ____

   ____

   c) Gas exchange – ____

   ____

10. What is a respiratory surface?

____

____

____

11. List four properties of a respiratory surface which aid in diffusion.

____

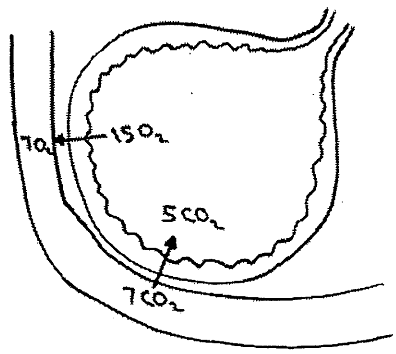
____

____

____

9

12. Using the diagram below showing the concentration of oxygen and carbon dioxide in the alveolar capillaries and air. Explain why carbon dioxide moves from the blood in the alveolar capillaries into the alveolar air, and why the oxygen moves from alveolar air into the alveolar capillaries.



13. Name and describe four mechanisms that maintain the diffusion gradient in the lungs.

14. Complete the following table contrasting the composition of inhaled air and exhaled air.

Component	**Inhaled Air**	**Exhaled Air**
Oxygen (%)		
Carbon Dioxide (%)		
Water Vapour (%)		

10

What do the lungs do? Video questions

1. The whole purpose of the respiratory system is to transport **oxygen/blood/carbon dioxide** to **blood/cells/carbon dioxide** via the **cells/oxygen/blood** (circle one)

2. ______________ goes in and _________________ goes out in the process of **respiration/photosynthesis**

3. List 3 parts of the respiratory system:

   ____

   ____

   ____

4. What is one main muscle used in breathing? _________________________________

5. When you breathe in (inhale), your lungs **contract/expand** because your muscles **contract/expand**

6. Number these in the order the air travels:

   Bronchi Nose and mouth Trachea Alveoli Bronchioles

7. Alveoli have tiny blood vessels around them called c____________________

8. Oxygen enters the blood and carbon dioxide leaves the blood through the process of d_____________

	Aerobic respiration	Anaerobic respiration
Reactants		
Products		
Chemical equation		
Energy gained (low or high)		

Respiratory System Worksheet

I. Complete the following statements by inserting one of the words below in the answer blanks.

cartilage	moisten	pressure	voice box
filter	nostrils	speak	warm
larynx	pharynx	vocal cords	

Air enters the nasal cavity of the respiratory system through the __________. The nasal cavity has several functions. The major functions are to ________________, ________________, and ________________ the incoming air. The passageway common to the digestive and respiratory systems, the ________________, is often called the throat; it connects the nasal cavity with the ______________ below. Reinforcement of the trachea with ________________ rings prevents its collapse during ________________ changes that occur during breathing. The larynx or __________________ is built from cartilage. Within the larynx are the ________________________, which vibrate with exhaled air and allow an individual to ________________.

Put the following in the correct order of how oxygen travels into the bloodstream?

Alveoli, trachea, bronchioles, capillaries, bronchi

What is the word equation for respiration?

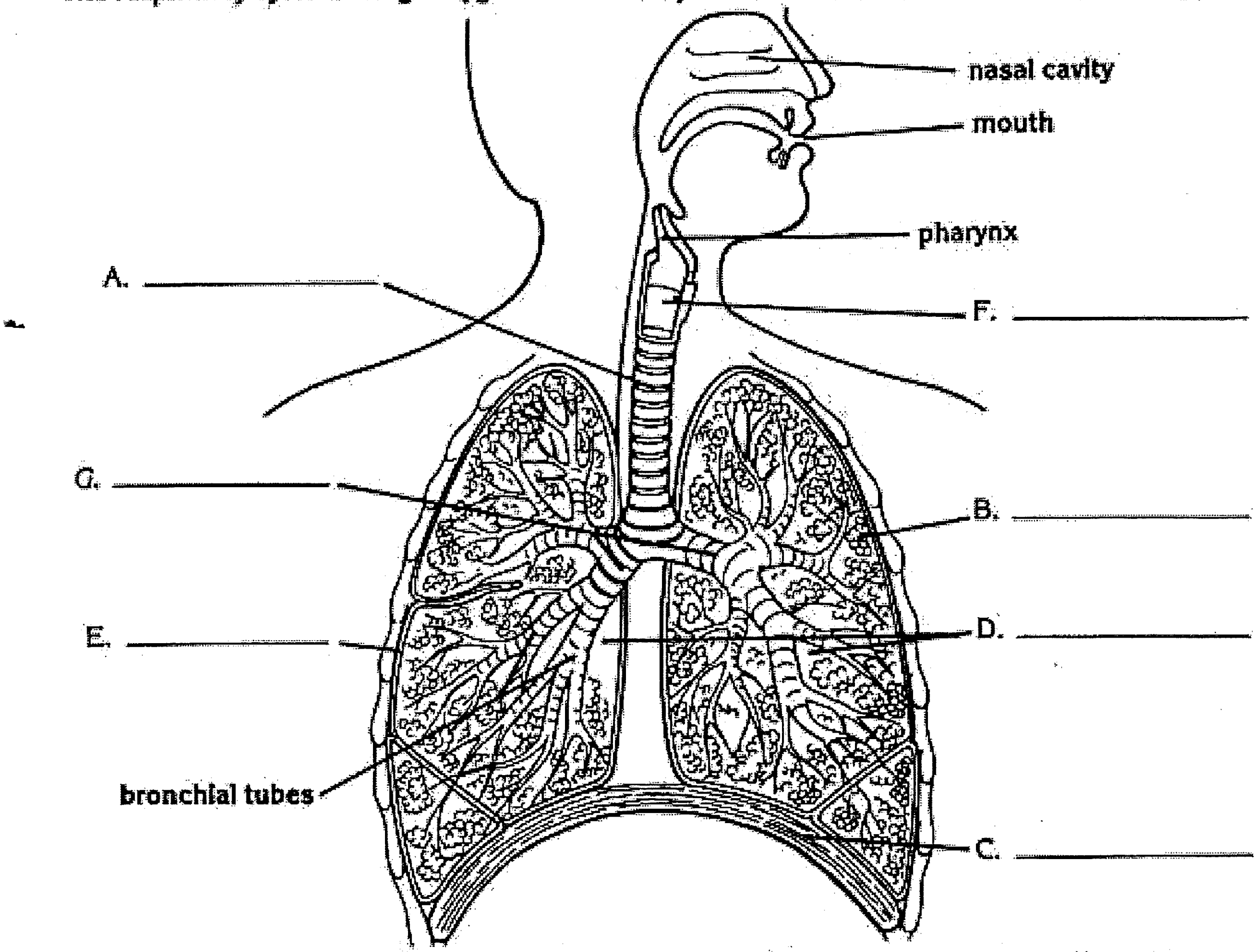
What are the reactants and products?

Explain why lungs work much better with alveoli rather than if they didn't have them.

Explain the process of breathing in and out. Use the words diaphragm, inhalation, exhalation, pressure, contract, relax, gas exchange.

The Respiratory System

The respiratory system brings oxygen into the body and removes carbon dioxide and other gases.



1. Study the diagram to correctly identify these parts of the respiratory system. Then use each answer to correctly label the diagram.

   A. the tube that connects the throat and the bronchial tubes ____________________

   B. the grape-like clusters of air sacs within the lungs ____________________

   C. the large band of muscle that controls the size of the chest cavity ____________________

   D. the two large lightweight respiratory organs of the body ____________________

   E. the outer membrane which covers the lungs ____________________

   F. the part of the respiratory system that helps us speak ____________________

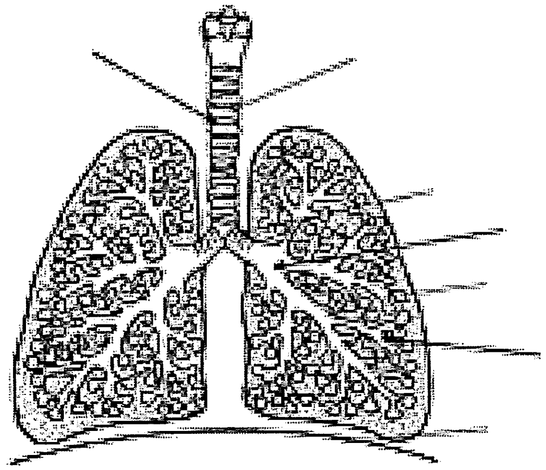
   G. the two branches of the windpipe ____________________

2. Write **True** if the statement is true. Write **False** is the statement is false.

   ____________ The teeth are an important part of the respiratory system.

The Structure of the Lungs

Label the diagram below using the words from the box.



TRACHEA
BRONCHUS
ALVEOLI
DIAPHRAM
CARTILAGE
BRONCHI
BRONCHIOLES

TRACHEA CARTILAGE BRONCHI BRONCHIOLES ALVEOLI
SWALLOW MUCUS EPIGLOTIS BRONCHUS DIFFUSES
CARBON DIOXIDE CELL OXYGEN CAPILLARIES ONE

The Respiratory Pathway

Air passes into the nasal cavity, where dust and Bacteria are filtered out by sticky M________. The air is drawn down the T____________ which is kept open by rings of C____________.

At the top of the Trachea is a small flap known as the E___________. When you S_________ it closes off the T____________ to prevent food going into the lungs. At the bottom the Trachea it divides into two B_________. Each B____________ subdivides again into a mass of fine branches known as B___________. These end in tiny hollow bags with bulb-like pockets called A__________ in which gases are exchanged. The walls of the Alveoli are only O_____ cell thick, and their outer surface is covered by a dense network of C______________. The blood flowing through these capillaries absorbs O__________ which D_________ through the alveoli and capillary walls. At the same time blood releases C__________ D____________ which diffuses in the opposite direction into the alveoli.

Respiratory and Circulatory System Disease Task **Name:** ______________

Choose one disease out of the following table (each student will have a different disease). Answer the following questions about the disease. You will be required to tell the rest of the class about your disease in an approx. 2min long verbal presentation (you may use powerpoint slides as well if you wish). This is not assessed, but will give everyone an idea of the different conditions that can affect us. (Note: you do not have to do referencing for this activity!)

Questions:

1. What causes it?
2. Which part of the respiratory/circulatory system is most affected/where does it occur?
3. What are some widespread effects it has on the body?
4. Are there ways to prevent it?

Respiratory System:	**Circulatory system:**
Asthma	Coronary artery disease
Pneumonia	Hypertension
Bronchitis	Cardiomyopathy
Emphysema	Peripheral artery disease
COPD - chronic obstructive pulmonary disease	Arrhythmia
Tuberculosis	Atherosclerosis
Sleep apnoea	Heart attack
Asbestosis	Deep Vein Thrombosis
Pulmonary Fibrosis	Aneurysm
	Stroke
	Varicose Veins

On the following page is a Cornell notes template to use only if you wish.

Title:

Key Points:

Notes:

Summary: